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Building an algorithmic organisation

A Case Study of the Transformation Journey of Flygresor.se

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Abstract

This study investigates how Flygresor.se, a Swedish meta-search platform for airline tickets. Managed to transform into a highly efficient, algorithm-driven organisation by leveraging data-driven decision making. Also, exploring how Flygresor managed to leverage this methodology to achieve remarkable financial resilience-maintaining 30% profit margins, surviving the COVID-19 pandemic without losses, and generating 15 MSEK revenue per employee by systematically replacing traditional management with automation and real-time data analytics.

This study also explores the limitations and potential of algorithmic management reaching beyond gig economy platforms, demonstrating its applicability to smaller firms. Its success and adoption of algorithmic management stems from four pillars: data utilisation, automation, user engagement, and social interaction. By collecting granular data at five-minute intervals (e.g., marketing spend, user behaviour, and revenue metrics), the company built systems like RelaxAd to automate marketing adjustments hourly. A modified flat organisational structure, reduced from 11 to 7 employees, decentralised decision-making and eliminated hierarchical roles. This study explores how a start-up spends a decade transitioning into an algorithmic organisation.

Keywords

Algorithmic management, algorithmic organisation, flygresor.se, organisation theory, data-driven decision making, AI, automation

Sammanfattning

Denna studie undersöker hur Flygresor.se, en svensk metasökningsplattform för flygbiljetter, lyckades omvandlas till en högpresterande, algoritmstyrd organisation genom att använda datadrivet beslutsfattande. Studien utforskar också hur Flygresor kunde utnyttja denna metodik för att uppnå anmärkningsvärd finansiell resiliens med 30% vinstmarginal, klara sig genom COVID-19-pandemin utan förluster och generera 15 MSEK i årlig intäkt per anställd – genom att systematiskt ersätta traditionell ledning med automatisering och realtidsdataanalys. Studien undersöker även begränsningarna och potentialen hos algoritmisk ledning bortom gig-ekonomiplattformar och visar dess tillämpbarhet för mindre företag.

Flygresors framgång och införande av algoritmisk ledning bygger på fyra pelare: dataavändning, automatisering, användarengagemang och social interaktion. Genom att samla detaljerad data med femminutersintervall (t.ex. marknadsföringsutgifter, användarbeteende och intäktsmått) byggde företaget system som RelaxAd för att automatiskt justera marknadsföring per timme. En modifierad platt organisationsstruktur, reducerad från 11 till 7 anställda, har decentraliserat beslutsfattandet och avskaffat hierarkiska roller.

Nyckelord

Algorimiskt styrning, algoritmisk organisation, flygresor.se, Organisationsteori, datadrivet beslutfattande, AI, automatisering

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Stockholm, July 2025
David Hu

Contents

1 Introduction **1**
1.1 Background . . . 1
1.2 Purpose . . . 2
1.3 Research Questions . . . 3
1.4 Research Methodology . . . 3
1.5 Scope & Delimitations . . . 3
1.6 Structure of the Thesis . . . 4

2 Research context: Flygresor.se **5**
2.1 Company introduction and description . . . 5
2.1.1 Organisation structure and operation principle . . . 6
2.1.2 Revenue management of advertising spending . . . 7
2.1.3 A decade of transformation . . . 9

3 Frame of reference **13**
3.1 The rise of algorithmic economy . . . 13
3.2 Algorithmic business model . . . 14
3.3 What is algorithmic management . . . 14
3.4 Dimensions and characteristics of AM . . . 15
3.4.1 Individual level . . . 15
3.4.2 Organisation level . . . 16
3.5 Defining algorithmic management . . . 17
3.5.1 Definition of algorithmic management in this thesis . . . 18
3.6 Designing algorithmic management systems . . . 18
3.6.1 Combining algorithmic management systems with human interaction . . . 19
3.7 Building an algorithmic organisation . . . 19
3.8 Summary . . . 20

4 Methods **23**
4.1 Research design . . . 23
4.1.1 Phase 1: Research context . . . 24
4.1.2 Phase 2: interview study . . . 24
4.2 Frame of reference . . . 25
4.3 Research context . . . 25

4.4 Collection of data . . . 25
4.5 Validation of results . . . 26

5 Results **27**
5.1 Interview findings . . . 27
5.1.1 Interview with the development team . . . 29
5.1.2 Key learnings . . . 31
5.2 Comparison case . . . 31
5.2.1 Data driven culture . . . 32

6 Building an algorithmic organisation **35**
6.1 Key constituents of an algorithmic organisation . . . 36
6.1.1 Data . . . 38
6.1.2 Automation . . . 39
6.1.3 User engagement . . . 41
6.1.4 Social interactions . . . 42

7 Discussion **45**
7.1 Key technical factors of an algorithmic organisation . . . 45
7.2 Constraints, limitations and difficulties associated with algorithmic organisation . . . 47
7.3 Shift in the study of algorithmic management . . . 48
7.3.1 Algorithmic management research within human resources . . . 48
7.4 Scaling an algorithmic organisation . . . 49
7.5 Limitations . . . 50

8 Conclusions and Future work **51**
8.1 Conclusions . . . 51
8.2 Future work . . . 52

References **53**

A List of interview candidates **1**

List of Figures

2.1 Organisation structure. *Note: the number within the parenthesises displays the number of full-time employees on each role, excluding the consultants.* 6
2.2 Daily work process of managing marketing spend using RelaxAd based on the internal dashboard and daily revenue. . 9
2.3 Pre-Covid activities after the acquisition by Etraveli group. . . 10
2.4 Pre-Covid activities after the acquisition by Etraveli group. . . 11

4.1 Descriptive flowchart of the research process and steps. 24

6.1 Foundation of the taxonomy with a bottom-up approach to algorithmic management. 37
6.2 Breakdown of the components contributing to the taxonomy, to identify the level of algorithmic management adopted. . . . 38
6.3 Examples of how a measuring structure can be built. 40

List of Tables

2.1 All of the different data types that Flygresor.se collects down to five minute intervals 8

5.1 Breakdown and summary of KPI structure 33

A.1 Interview candidates 1

List of acronyms and abbreviations

ADM	Algorithmic decision-making
AI	Artificial Intelligence
AM	Algorithmic management
AMS	Algorithmic management systems
ARR	Annual recurring revenue
DDD	Data driven decision making
GUI	Graphical user interface
HRM	Human resource management
KPI	Key performance indicators
MMM	Marketing mix model
RM	Revenue management
ROAS	Return on ads spent

1. Introduction

This chapter presents the background of the thesis project, along with its purpose, research questions, and delimitations.

1.1 Background

The rapid development of digital technologies, particularly in big data and Artificial Intelligence (AI), has transformed organisational management and decision-making processes. More and more firms are starting to use data-driven methods to gain a better understanding of their business and make more informed decisions. In this new context, productivity and efficiency are at the heart of this transition. The introduction of data-driven organisations and Data driven decision making (DDD) has been pivotal for some firms. Research conducted by LaValle et al. (2010) has shown that by using data-driven methods to differentiate their offerings in the market, firms have twice the chance of being top performers. Insights like this motivate companies to build systems and automation leveraging these types of technologies, empowering their decision-making process by using data and algorithms. A clear benefit to data-driven action is that it allows firms to gain competitive advantages and increase their return on investment (Phillips-Wren and Hoskisson, 2015).

The integration of big data and data-driven methods into organisational design can further amplify these effects. This is achieved by leveraging large amounts of data, which, with the help of algorithms, can optimise resource allocation, improve operational efficiency, and even replace workers with software (Schildt, 2017). This can be done by the implementation of Algorithmic management systems (AMS). During the rise of on-demand platform solutions like Uber and Lyft, a new challenge emerged: managing large fleets of workers and creating dynamic solutions for end-users. Since these platforms operate in a two-sided market, they must satisfy both parties. By acting as a middleman with multiple stakeholders involved in each transaction, complexity increases, and the number of variables also increases. This will also generate more data points and add more complexity to the transaction. In the context of analysing

and processing data with multiple variables, algorithms are particularly well-suited and become an effective solution. Uber and Lyft were among the first to adopt this management style because of their vast numbers of drivers. By using software, they were able to manage large numbers of workers and implement dynamic on-demand pricing, giving them a significant advantage over traditional taxi firms (Lee et al., 2015).

This is where the case of Flygresor.se is particularly interesting. Flygresor.se is a flight meta site founded in 2004 and was acquired by Etraveli Group in 2014. Since the acquisition, Flygresor.se has managed to maintain its start-up culture while growing the business and continuously lowering its overhead costs. Originally starting with six employees at the time of acquisition, the firm grew to eleven employees and, as of 2025, has scaled back to eight employees. The firm has an annual recurring revenue of approximately 120 MSEK, equating to roughly 15 MSEK per employee. Despite these changes, Flygresor.se has maintained a margin of 30% and survived the COVID-19 pandemic without reporting any losses during this entire period. A remarkable feat in a sector plagued by revenue losses, bankruptcies, and lay-offs. Flygresor.se experienced none of these issues. This article aims to understand and investigate how data can be used as a powerful and actionable tool for management.

In this context, this paper explores the intersection of data-driven organisations and algorithmic management, focusing on technologies and software that can enhance operational efficiency and revenue management. Taking a deep dive into Flygresor.se's transformation journey. Presenting their learnings and failures to become more resilient.

1.2 Purpose

Algorithmic management is still an early concept. There is thus no established framework and structure on how to build an algorithmic organisation. Even firms that work this way don't know the concept of an Algorithmic management (AM) organisation. Studies today tend to focus on people management and how AM can contribute to firms with a large number of employees, but does not attend to its potential application in firms with a small number of employees. The purpose of this study is to analyse Flygresor.se's transition journey from a startup to a high-earning company within their field

by developing algorithms and software catering to their needs. Allowing the CEO to manage the firm through a systematic approach and apply data-driven decision-making on an hourly basis.

1.3 Research Questions

- What key technical- and organisational factors enable successful transformation of start-ups into algorithmic-driven organisations?
- What are the key constraints, limitations, and difficulties associated with algorithmic management in organisations, and what strategies can be employed to mitigate these challenges?

1.4 Research Methodology

This study consists of two parts. These are; a background study of Flygresor.se and a qualitative study of the current employees of Flygresor.se. The aim of the background study is to build an understanding of how they have operated in the past 10 years. This study is conducted via unstructured interviews at their offices. The main study is done with semi-structured interviews to build upon the background study.

1.5 Scope & Delimitations

This paper explores these concepts on an organisational and management level and in doing so excludes the analysis of statistical models and algorithmic software development. But focus on creating the necessary organisation structure and systems requirements for the development of such software. This study will not interview or contact external stakeholders and will only analyse Flygresor.se's internal software development process and what critical requirements are needed to achieve the current operation systems. In previous research regarding AM discussion regarding surveillance capitalism, and also the issue of data policy is heavily addressed as an issue (Onsrud and Campbell, 2020). While this topic is important, this study will only focus on the individual and organisation level of the topic,c excluding the macro effect of such implementation.

1.6 Structure of the Thesis

The thesis will start with the pre-study of Flygresor.se in chapter 2 followed up with the Frame of reference 3 giving a overview and background on the topic of AM and how do design for AM. Following up with the research design and the structure of the study. In chapter 5 a comprehensive review of Flygresor.se will be presented with a detailed time and description of their transformation journey. Following the findings a recommendation guideline 6 will be presented as recommendation on how to build an AM focused organisation. This is continued with the Discussion 7 and Conclusion 8.

2. Research context: Flygresor.se

A background study was conducted to explore and analyse the organisational structure of Flygresor.se. This was achieved through interviews and on-site observations during their workflow. Most of the interviews were unstructured and were conducted ad hoc during the study period.

2.1 Company introduction and description

Flygresor.se is currently one of the largest flight comparison sites in Sweden, capturing approximately 50% of the Swedish airline meta market. Meta platforms are defined as aggregators of multiple streams of information and services; meta systems do not own or produce any products. A similar service in Sweden would be Pricerunner. Key competitors to Flygresor.se are Skyscanner and Momondo/Kayak, thus excluding firms such as Booking.com and traditional flight ticket sales channels. Flygresor.se functions as the middleman between travel agencies, airlines and the end customers. Traditionally, travel agencies would pre-buy plane tickets in advance and bundle these offerings with their charter packages, along with other services they provide. To prevent losses in the case of low demand for charter packages, these tickets are sold on meta platforms like Flygresor.se. Flygresor.se brokers these tickets to the end consumer, sometimes at a lower price. Because of the competitive landscape of airline tickets and customers' prioritisation of price, price comparison sites own a significant part of the airline ticket industry, as they allow customers to compare prices from multiple sources before making a decision. This is how Flygresor.se generates revenue: by redirecting end customers to purchase tickets from travel agencies and taking a broker fee.

As of 2024, Flygresor.se has a recurring revenue of 120 MSEK and eight full-time employees. The company has been in operation for over 15 years. This thesis will primarily focus on the past decade of their operations. Over this

period, Flygresor.se has systematically eliminated multiple management roles (see Figure 2.4) and now operates with a high degree of automation.

2.1.1 Organisation structure and operation principle

Flygresor.se operates with a modified flat organisation. While there are hierarchies within the firms. The communication and decision-making are decentralised to the project members. Figure 2.1 displays the breakdown of the organisation structure. This organisation consists of a total of 8 people. Working directly under the CEO. The CEO's role in Flygresor.se might differ from traditional CEO roles. In this organisation, the CEO acts as the marketing manager, product owner and manages the entire business operations. The CTO works directly with the CEO and acts on the directive from CEO. As mentioned before, these directives are often vision-based. The CTO is given full autonomy on how the goals are achieved and task distribution.

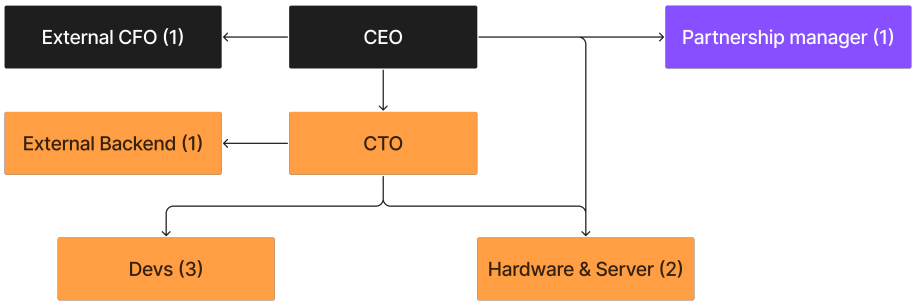


Figure 2.1: Organisation structure. *Note: the number within the parenthesises displays the number of full-time employees on each role, excluding the consultants.*

The operating principles at Flygresor.se play a key role in its development. A key lesson from the CEO is *"Get your house in order to be able to seize the opportunity when it occurs."* Even the best strategies will fail if the core business is not robust enough to withstand pressure. One of the operating principles is the shift from strategy to vision and tactics. This is done by splitting the vision into short-term (3–12 months) and long-term (12 months or more) tactical plans. Before starting a task, each employee should ask themselves whether the activity aligns with the vision. By enforcing this,

employees are free to act as they see fit, as long as it aligns with the company vision.

Transparency is a key driver of success at Flygresor.se. Every employee has access to the company dashboard, allowing everyone to track the company's performance, whether good or bad, in real time. This eliminates information asymmetry between management and employees.

Another factor contributing to their success is their generous work-from-home policy and flexible working hours, which provide employees with a high degree of freedom in where and how they work. Employees are given both autonomy and responsibility in decision-making and are encouraged to contribute ideas to help shape the company's products.

2.1.2 Revenue management of advertising spending

Flygresor.se's main cost driver is their marketing expenses, which also serve as their primary customer acquisition channel. To put this into perspective, they spend approximately 75 MSEK annually on digital marketing, which corresponds to approximately 69% of their total revenue. They currently advertise on eight different platforms, including Google, Facebook, TikTok, Snapchat, and others.

There are multiple purchasing methods when it comes to advertising, with the two main ones being cost per click and cost per action. This creates a significant challenge for the daily management of their advertising spend. Traditionally, an operation of this size would be outsourced to a media agency. However, in Flygresor.se's case, they manage this in-house. This is accomplished by isolating the ”small data” needed from the marketing platforms and then aggregating it into an algorithmic software called RelaxAd.The ”small data” is provided by an internal dashboard that is built in-house. The ”small data” can be found in table 2.1; these are the data points that Flygresor.se tracks at 5-minute intervals.

Table 2.1: All of the different data types that Flygresor.se collects down to five minute intervals

Earnings	**Costs**	**Brand**	**Traffic**
Daily profits	Overhead costs	Brand awareness	Users per day
Net Revenue	Operating working capital	Top of mind	Meta all market share
Net Revenue Actual	Click cost	In mind	Click-out per hour
Net Revenue Budget	Ad spend digital		Searches per hour
Net Revenue Difference	Other costs		Click rate
Time of profitability	Marketing costs		Clicks
EBITDA Budget			
EBITDA Actual			
EBITDA Difference			

By aggregating this data to one platform it can calculate Return on ads spent (ROAS) and based on this Key performance indicators (KPI) the CEO can make hourly adjustments to the marketing spend. The CEO daily operation working as displayed in figure 2.2. RelaxAd is also feeding marketing data back into the dashboard. All of the marketing decision and adjustments are done in RelaxAd. The internal dashboard only acts as a aggregate of information. Within the systems of RelaxAd Algorithmic decision-making (ADM) is involved in the daily operation. The software acts as an interface, and it can also give suggestion regarding how to adjust marketing spend and strategy.

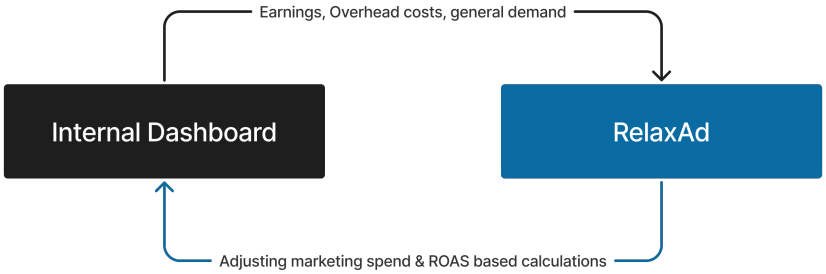


Figure 2.2: Daily work process of managing marketing spend using RelaxAd based on the internal dashboard and daily revenue.

2.1.3 A decade of transformation

Since the acquisition by Etraveli group in 2014, the company have gone through a transformation journey from a startup to a stable business. This period will be divided into two sections these are the pre-COVID era and the COVID era. This separation is done because of the impact COVID had on the airline business. The algorithmic development was started due to the realisation of them losing market leadership in 2016, which prompted a year of extensive restructuring and the creation of Marketing mix model (MMM) to build measurement pipelines such as daily traffic, marketing spend, revenue and prognoses of the long-term effect for the actions taken today. All these factors were taken into consideration when building the MMM model. See Figure 2.3 for a comprehensive timeline before COVID.

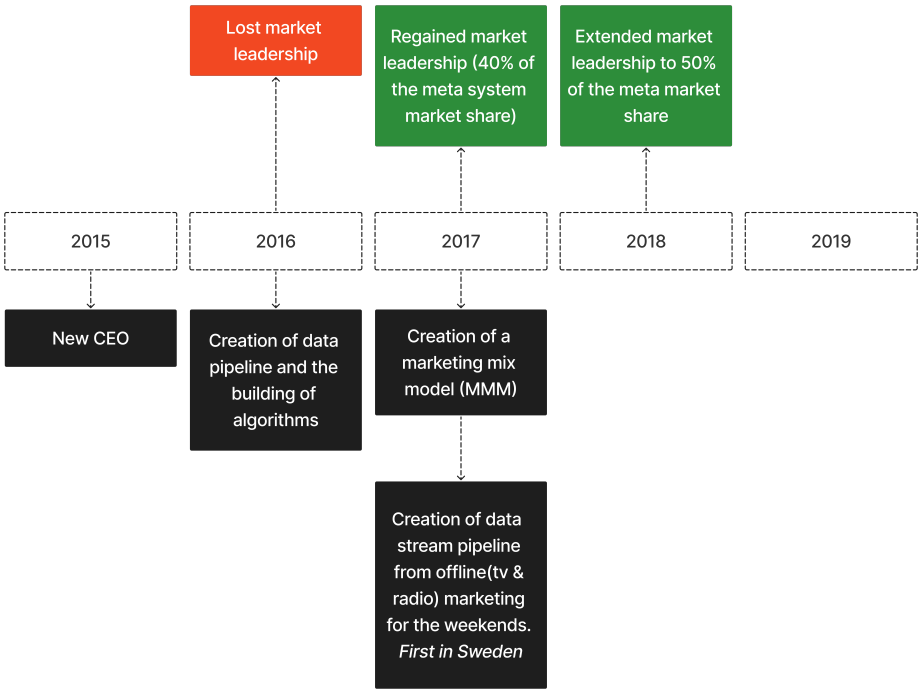


Figure 2.3: Pre-Covid activities after the acquisition by Etraveli group.

Shortly after the acquisition, a new CEO took over the firm and in 2016, they noticed that they had lost market leadership. This was mainly because they lacked the necessary pipeline to measure the traffic data; this was also the start of algorithmic control. Market lead was regained after one year of restructuring and building new ways to measure and analyse the daily data. This was combined with an aggressive marketing strategy to regain the market share. Due to the usage of multi-channel marketing, a MMM was created to aid in the effort of understanding which of the channels were most profitable. By leveraging this knowledge, the marketing officer could be deliberate in the deployment of marketing funds. Thus, yielding better results. MMM are known to be challenging to build. Meaning that to build this MMM Flygresor.se needed to rebuild and upgrade their current data pipelines and create a more detailed way of measuring and collecting data. This was difficult and took multiple years to get it right.

During this period in 2017, offline media were still a popular choice for advertising. These were TV and radio, but the issue was that the data for the

number of viewers of a TV ad was delayed by eight days during the weekends. Which was an issue because this would mean that the data for the MMM was too old. To combat this, they created a new way of measuring data and a new pipeline together with the broadcasting channels, which allowed them to receive the weekend data on Monday instead of Friday. This was the first in its kind at that time. By collaborating with different publishers and suppliers of advertising, it allowed Flygresor.se to build an algorithmic model that could learn from previous data and create a prognosis of future earnings, which allowed the marketing officer to act in real time, dependent on the model. Thus, gaining the most value from the marketing budget.

COVID lockdowns and regulations put a strain on the airline and travel sectors. During this time, many services within this space were struggling. But for Flygresor.se, they were surviving without any issues. Because of their automation systems, they were able to turn off and lower all of their marketing spend and all of the costs related to this. This was done by analysing the macroeconomic trends and issues, but a driving factor for this is because of their algorithmic system and dashboard. This allowed them to regulate their operating cost at will at any time of the day.

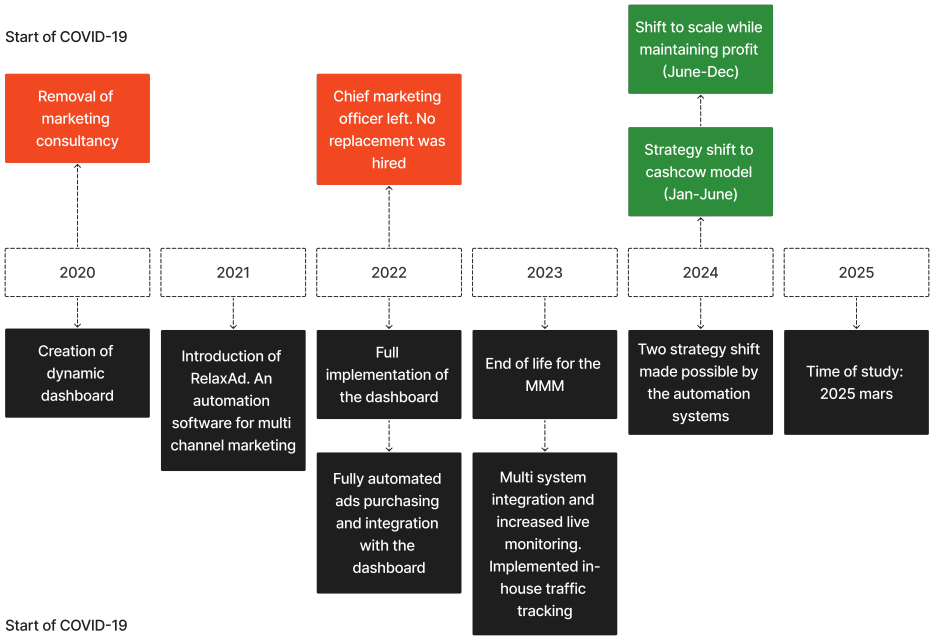


Figure 2.4: Pre-Covid activities after the acquisition by Etraveli group.

3. Frame of reference

3.1 The rise of algorithmic economy

Revenue management (RM) was first developed for the airline industry with the idea of demand-based pricing on plane tickets based on availability. This idea of systematically calculating the demand and price based on this information was proposed by Littlewood (2005) in 1972. This started the first iteration of the term RM. The most widely adopted RM models are quantity-based approaches that are based on the number of passengers booked and the number of days left before the take-off date. By doing so, this allowed the airlines to generate significantly more revenue and motivated the buyers to buy tickets long ahead of time. By adopting this methodology and collecting customer data, it was possible to map out high-yield and low-yield customers based on customer profiles and purchase patterns. These data points could be age, number of travels and type of seat. This approach also sparked the idea of collecting a large amount of consumer data and mapping out consumer behaviours.

RM has played a significant role in markets such as e-commerce and digital ad networks. The increased level of digitalisation allows for increased flow of information and monitoring possibilities. These innovations, coupled with the development of platforms such as Google Ads, Facebook marketplace, Instagram ads and so on, have created a shift from traditional brick and mortar stores to online marketplaces instead. By transitioning to digital marketplaces, the demand for live monitoring increased, and real-time product tracking on inventory and customer traffic created a need for a new model. This model is called the GvR model by Gallego and Ryzin (1994) developed in 1994 before the digitalisation era, is still viable in today's market. Compared to the Littlewood (2005) model, which only accounted for static single resource inputs for the optimisation of value, the GvR model incorporates dynamic pricing and demand management over time, making it better suited for environments where both demand and available inventory can fluctuate rapidly. This adaptability is crucial in digital marketplaces, where consumer

behaviour and inventory levels can change in real time. The GvR model allows for continuous adjustment of prices and allocation of resources, optimising revenue based on current market conditions rather than relying on fixed, pre-determined values. As a result, businesses can respond more effectively to shifts in demand, maximise resource utilisation, and ultimately improve profitability in the fast-paced digital economy.

After the growth of online retailing and online advertising, the usage of RM continued into ride-hailing platforms. These industries act a brokers of customer acquisition for the buyer. Shifting from demand management to conversation optimisation on how to maximise the customers' spend. Digital advertising and social media platforms allowed for 24/7 display of ads and monitoring of behaviours. Similarly to Littlewood's model, it generates user profiles and optimises the return based on individual purchasing behaviour. Thus, introducing AM within the ride-hailing business.

3.2 Algorithmic business model

The algorithmic approach to business model is creating a pricing model based on the market's willingness to pay at a certain moment. Small increments and changes to pricing in large quantities can generate significant gains. What was once linear is now dynamic. For example, will you be willing to pay more for a cold can of Coke when it is 30 degrees outside? It is this type of question that is constantly being asked and optimised in an algorithmic run business. By processing the consumer data and mapping out their behaviour, the firms can, in theory, increase coke prices based on key factors, and the consumer will still purchase the can. By doing so, leveraging the algorithmic approach to business models. (Seele et al., 2021; Dolgui and Proth, 2010).

3.3 What is algorithmic management

AM was first introduced by Lee et al. (2015) during their studies of Uber and Lyft. It is a systematic approach regarding the management of personnel in a two-sided market where the company acts as a broker for a service or product. This methodology explores the implementation and usability of automating traditional human interactions with systems and software. This allows for increased efficiency. AM emerged during the rise of sharing and gig economy

platforms with the focus on peer-to-peer transactions. Because of the large scale of peer-to-peer transactions, it is not possible to have a human worker manage over 1,000 peer-to-peer actions with tasks such as matching drivers with clients. For platforms such as Uber, it uses algorithms to accomplish such tasks, automatically assigning tasks, daily planning and dynamic pricing of services. It also enables live tracking and introduces award and punishment systems based on performance without the need for human interaction.

Companies like Uber and Lyft use AM to automate worker-to-customer coordination, replacing traditional human mediators with software-driven algorithms. This is regulated by the demand and supply; in their case, the geographic information is also taken into consideration (Möhlmann et al., 2021). So, by using customer and worker (driver) data, it can predict on if the driver's likelihood of accepting a fare depends on the distance and also the payout. This is the same when it comes to the customer. This kind of thinking has revolutionised the taxi business and how fares are being priced.

3.4 Dimensions and characteristics of AM

Algorithmic management is a complex topic with multiple dimensions and mechanisms. Generally, it is used for worker management and used within Human resource management (HRM). For example, algorithmic controlling is a way of using algorithms to supervise workers' performance in real time and making sure that it is aligned with the organisation's strategy and vision (Möhlmann et al., 2021). Dividing the dimensions into the individual level and the organisation level allows for a more nuanced analysis of this issue.

3.4.1 Individual level

Implementation of AM allows for a high level of autonomy for the workers. Because of the lack of contact with a supervisor, the actions and task assignments are distributed and logged automatically into the system. Most often, the compensation and pay will also be based on the logged data. This offers a high degree of flexibility, autonomy, and task variation (Wood and Lehdonvirta, 2021; Wood, Graham, et al., 2019). By reducing human interactions in the daily operations, information asymmetry will occur. In the case of Uber and Lyft, the drivers are automatically assigned tasks which they

can accept or decline, this is similar to working for a black box. The drivers lack knowledge of the task assignments. Designs like these allow the platform owners to control the drivers based on their behaviours and ”nudge” them in a certain direction without the drivers even knowing it (Wood and Lehdonvirta, 2021).

In an AM firm with high employee turnover and easy sign-ups, job security for employees is reduced.. This is prevalent in the gig economy and the food delivery platforms. The recruitment process is automated, which allows the platforms to process a large number of applicants in a short amount of time. By using AMS for recruitment, it will reduce the downtime and always present a healthy array of work opportunities for the workers. While this is positive, it also means that the platform implements automation systems for worker performance measurement and allows for automatic termination (Möhlmann et al., 2021). This level of disconnect with co-workers may also lead to social disconnection and the loss of a sense of belonging, which in turn can cause overwork and mental issues.

3.4.2 Organisation level

By implementing AM as a methodology and practice within the organisation. Its needs will also shift based on the organisation design and operating principles. With the more efficient tracking and monitoring, middle management could be decreased in headcount in favour of ADM using software instead. Building machine learning models that can be trained by managers, thus unloading their daily need for reporting and processing of data. This shift also entails a different need of talent and knowledge. Shifting towards algorithmic trainers, explainers and IT personnel. By doing so, introducing a leaner middle management and the birth of new roles and a new organisation structure with a lean middle level and top level, creating a pyramid-like structure in headcount (Jarrahi et al., 2021).

By restructuring the organisation and reassigning power to AMS will allow the managers issue of data overload, in the sense of work arrangement, performance reporting and CV screening for recruits. All of these tasks can now be automated by algorithms.

Thereby improving the management quality (Jarrahi et al., 2021). But

the imbalance in data accessibility between managers and workers could also worsen the information asymmetries. Managers can create ”nudges” in the form of penalties or rewards based on worker performance in daily operations (Mateescu and Nguyen, 2019). So, the implementation of AM relinquishes operational control to managers, removing the mundane tasks of data collection and data cleansing to algorithms instead. It can also boost the accuracy of decision making on an organisation level, but it can also increase information asymmetry and a further level of disconnect from its workers. This means that managers in AM organisation need to develop other skills within the social technical space, skills such as social intelligence and tacit understanding (Jarrahi et al., 2021).

3.5 Defining algorithmic management

There is no universal definition of AM. The term was first used by Lee et al. (2015), who defined AM as *”-software algorithms that assume managerial functions and surrounding institutional devices that support algorithms in practice algorithmic management.”* This was a rather general definition of the topic. Mateescu and Nguyen (2019) defines AM as using technological tools and techniques to manage the workforce, relying on data gathering and analysis to automate the decision-making process. This definition is adopted from AM usage in the gig and sharing economy. Similarly, Wood, Graham, et al. (2019) defines it as a form of digital control, which is AM that allows for a platform-based rating and reputation system.

Duggan et al. (2020) defines AM as a system of controls where a self-learning algorithm can make managerial decisions affecting labour, thereby limiting human involvement in the matter. The algorithm is adaptive and can engage with the worker and act based on the individual’s data. This view is also adopted by Möhlmann et al. (2021), who emphasise the collection of large-scale data on platform-based services, which allows for the training of algorithms and enables them to carry out the coordination and control of workers. These are tasks that were traditionally performed by managers. Tomprou and Lee (2022) defines it as the automation of traditional managerial practices, AM can be used to automate tasks responsible for middle or upper management. Looking at previous research within the area.

Past literature tends to focus on the reduction of managerial roles within the

”gig” economy and misses the application of AM in the traditional workspace. Some also define AM as a method of managing workers for short-term contracts (Parent-Rocheleau and Parker, 2022). This is more prevalent in the gig economy by using this mechanism, which allows firms to control their freelancers in unconventional ways (Jarrahi et al., 2021).

3.5.1 Definition of algorithmic management in this thesis

Looking past the previous definition of AM for this paper AM will be defined as: a *set of systems and software with a self learning algorithmic interface. That empowers the users to control and manage the business operations, with real-time data allowing for real-time strategic shifts.* While the areas of gig and sharing economy have been heavily researched and explored. Taking a step back into the exploration of RM and how AM can be used to further develop this concept of maximising efficiency and implementing ADM while maintaining the human engagement.

3.6 Designing algorithmic management systems

AMS acts as the interface between the human aspect and computer-generated content. In an algorithmic organisation, the design of such an interface is challenging because this creates the same issue in a double-sided market. The goal of the interface should be to create a joint decision-making and problem-solving solution where both sides are included. By introducing automation and AI aided systems, humans are more distanced from decision making, and the algorithms act instead as the basis for knowledge and decision making because of their ability to trace data sources, collect data and perform data analysis. In contrast to human decision making, where the decisions are made based on experience, intuition and context (Klein and Sullivan, 2001). Algorithmic decisions are made based on statistical models and have the ability of faster iteration and more accurate data precision (Bader and Kaiser, 2019).

There are multiple types of algorithms, which can be categorised into three categories: descriptive, predictive and prescriptive algorithms. Descriptive

algorithms are used for simple tasks. Their main function is to analyse data by extracting, sorting and summarising the results and also conducting simple statists analysis and the input. Predictive algorithmics are used to predict events and actions based on the input data and using probability to do so. The last sort is prescriptive algorithms, which build on predictive algorithms by using simulations and scenario-based techniques to propose relevant recommendations (Parent-Rocheleau and Parker, 2022). Depending on the application, the selection of algorithms can differ, and it is important to understand which level is necessary for the application. Associated with the type of algorithms is also the degree of automation.

3.6.1 Combining algorithmic management systems with human interaction

AMS should act as a complement to their human counterpart rather than simply replacing the humans. By building an interface, it can be seen as an assembly of human and algorithmic intelligence working in symbiosis rather than one eliminating the other. This level of engagement is built on the interface level, since different types of media, and Graphical user interface (GUI) have different effects on an interface that requires more or less user involvement (Bader and Kaiser, 2019).

The goal should be to strive for equal involvement from both the user and the algorithms. If a user is presented with a firm recommendation without any explanation and relevant data, it would cause a low level of engagement since it is too easy for the user to just click accept and move on. Instead, the interface should present the user with the relevant data needed for decision making. In this design, it will engage the user in the decision-making process and combine the data analysis with previous experience from the user (Bader and Kaiser, 2019). This also empowers the users to take ownership of the decisions.

3.7 Building an algorithmic organisation

The foundation of an algorithmic organisation is built on data streams and the efficiencies of these data pipelines. Along with an understanding of what kind of data should be processed. Since building large data handling systems and the integration of big data could be costly. Job design in an AM organisation

is different then a traditional organisation. Since algorithms are just rules set by the user, the managers need to understand how to set these parameters to effectively use the interface. Also, with limited access to their managers and the high level of autonomy among employees, the manager requires tacit knowledge and also social and technical skills. Companies

ADM and AM increase the firm's dynamic capabilities, allowing for better adaptation to the external environment. This will also entail changes in organisational structure and design. While changes can be beneficial, the frequency of change should be low. Since rapid change of organisation structure or organisation function could lead to uncertainties with the employees and also create instability within the firms (Pisano et al., 1997).

AM gives more power to the managers since it will allow for closer monitoring and overcome cognitive limitations regarding data processing and management. By using complex algorithms, it allows for the manager to set multiple measurement parameters, while traditionally this would require the manager to spend a large amount of time to comply with the data, clean the data and perform analysis to then make a decision based on this report. But power imbalances can have a wide social impact on the work and also affect the company dynamics (Jarrahi et al., 2021).

AM organisations are more nimble and agile, allowing for quick changes dependent on the external environment. Rapid changes can be good and bad depending on the frequency. This means the need for continuous education of AMS and how to effectively use them. These types of education could also put strain on the firm's business operations. Since more time will be spent on education rather than revenue-generating operations.

3.8 Summary

AM and algorithmic business models are transforming multiple industries by leveraging data, and by adapting DDD the firm can optimise efficiency and streamline operations. Building on the foundation from RM, which emerged from the airline industry as a way of profit maximising and building a demand-based pricing model in the case of the Littlewood (2005). AM introduces a dynamic, real-time systems that automate managerial tasks. The early RM model was focused on static inputs, which proved inefficient for the usage

of large datasets and uncertainties. The GvR model by Gallego and Ryzin (1994) interoperates dynamic pricing and diminishing returns, making them adaptable to modern digital marketplaces.

AM emerged prominently in gig and sharing economies, where platforms like Uber, Foodora, Lyft and many more use algorithms to automate worker-to-customer coordination. These systems can replace traditional human workers with a more software-driven approach for task assignment and management. While AM enhances efficiency, it also raises concerns about transparency and information asymmetry due to the ”black box” effect. On an individual level, AM offers autonomy and flexibility but reduces transparency and social connections. Automated recruitment process streamlines hiring, but can lead to worse job security. On an organisational level, AM reshapes structures by reducing the middle management and relying on algorithm trainers and education for existing middle management.

Building an algorithmic organisation requires robust data pipelines as a foundation. Middle management need to develop technical expertise and skills to be able to set algorithm parameters and adjustment during the daily operations. Managers need to further develop social technical skills set guide employees that are operating in autonomous roles. This balance is need to ensure that the algorithmic systems are used effectively while maintaining human engagement within the organisation.

It is also a challenge to define AM since it is still a rather new concept and theory authors tend to focus on emphasising its role in automating workforce management through a self learning algorithm. This paper aims to challenge this definition and contributes to how rather the why. Thus defining AM as: *a set of systems and software with a self learning algorithmic interface. That empowers the users to control and manage the business operations, with real-time data allowing for real-time strategic shifts.* Exploring new avenues on the development of AMS and how it can incorporate human intelligence rather than just replacing humans. Creating ownership and belonging the the process. So, what key technical- and organisational factors enable successful transformation of start-ups into algorithmic-driven organisations? And what are the key constraints, limitations, and difficulties associated with algorithmic management in organisations, and what strategies can be employed to mitigate these challenges?

4. Methods

4.1 Research design

To build on existing theory and also expand on its practical implications and ways of implementation. The selection of the case company was guided by the extreme selection technique (Gerring, 2016). The methodologies fit well because of their ability to find new approaches by deriving insight from significant high values. While Flygresor.se have a small workforce and does not score high on AM compared to firms like Uber regarding their usage of employee management. The adoption of AM allowed the firm to downsize while maintaining performance. Studying human involvement in the regard of algorithmic decision making and practical level will require an in-depth understanding and analysis of the technologies in question and their implementation (Günther et al., 2017).

The research design stems from curiosity about analysing and discovering an extreme case of how a company can operate at an extremely high level of efficiency with a minimum number of employees in a highly competitive market while still maintaining profitability during multiple crises. The research will consist of two phases. See figure 4.1 for a full breakdown of the research process.



Figure 4.1: Descriptive flowchart of the research process and steps.

4.1.1 Phase 1: Research context

During this research phase the primary goal is to gain an understanding of how Flygresor.se operates and conduct an inductive study process to explore potentials and possibilities of the firm. These study are conducted in an unstructured manner by conducting casual conversation with employees and having general discussion regarding different topic. Adopting the inductive approach without any goal of question in mind.

4.1.2 Phase 2: interview study

During this phase an comprehensive study regarding Flygresor.se will be conducted with the goal of understanding the different pitfall and knowledge

that the firm have obtain during the past decade. A separate case study will be conducted on a compression firm as well to gain a wider understanding of the topic. See section 4.4 for an detailed description of the data collection process.

4.2 Frame of reference

Previous theoretical studies and frameworks were examined and analysed to create a good understanding of the area. By researching previous studies within a data-driven organisation, it can lay the foundation and understanding for the reader, along with strengthening the argument using academic studies. The search words used are ”algorithmic management”, ”Data-driven decision making”, ”innovation management”, ”Business model”, and ”Double-sided market & pricing”.

4.3 Research context

A background study was conducted with the Flygresor.se management team. During this period, the researchers were sitting next to the CTO and the CEO of Flygresor.se. During this time, multiple unstructured interviews and discussions were conducted ad hoc. This process allowed for an unstructured way of iteratively collecting past information and gained a better understanding of the firm. The researcher was also working from Flygresor.se offices to observe the work process and workflows in action. This, combined with structured interviews regarding their timeline and past experience, has created a holistic view of their transformation journey.

4.4 Collection of data

In the interview study, a total of fifteen (15) interviews were conducted. These interviews are conducted in a semi-structured approach. The majority of the interviews were held in person in the office. All interviews are recorded and transcribed. After transcribing the interviews, a thematic analysis will be conducted. The interview result will be clustered, edited and reviewed into themes. This analysis is done in MATLAB and exported to an Excel file. Besides the semi-structured interview, daily discussions and conversations are conducted during this study. See Appendix A for the interviewees.

4.5 Validation of results

To gain a holistic perspective of Flygresor.se operations, all of the employees were interviewed, and after the interview, a follow-up session was conducted with each individual to ensure that its content is correct. Further validation is also made by using the thematic approach for analysis. While the thematic approach allows for the detection of correlation. It could also lead to false-positive results. Such as a lack of standardisation and subjective interpretation of the questions. To combat this, two test interviews were conducted with outside personnel to ensure reduced the possibility of subjective interpretation of the questions was reduced. A limitation of this study is the lack of comparison cases. Since this study is conducted with Flygresor.se as its primary subject. A comparison study of multiple firms is beneficial to confirm the findings of this study.

5. Results

This chapter presents to study result from the two different case studies and qualitative research regarding the firms and its findings.

5.1 Interview findings

There is a heavy emphasis on autonomy and freedom for innovation found among all of the interview candidates. While executive roles exist, they exist only on paper. The CTO sees himself as a senior developer rather than a manager. This is also the case for the CEO. They adopt an open landscape office, which is common in Sweden, but what stands out is the lack of meetings. ”Meetings” are started just by calling out to each other and starting a discussion in place, and involving more people if necessary. The team have an acute understanding of their vision and goals, and all of the employees have a high level of autonomy and freedom. It is almost like the team at Flygresor.se is working in symbiosis, and there is a common mind running the firm.

Another aspect of the success of Flygresor.se is the first mover advantage. This was also mentioned multiple times. *”We were first and had 50% market share at the start”.* - Senior hardware engineer. This meant that Flygresor.se had a head start, but maintaining the market leadership proved to be difficult at the start. Flygresor.se lost their market leadership (See Figure 2.3) in 2016, which led to the advancement of their marketing efforts and also sparked the idea of building an algorithmic approach to their marketing and financial data.

Not long after the change in management and the arrival of the new CEO, a new controller was also hired to manage the financial issues. In the beginning, it lacked structure, and the information was always delayed. This meant that the accounting was a mess, and also a lot of delays in costs and missed invoices. The previous controller mitigated this by taking a lot of notes and having a separate Excel sheet to manage this chaos. This can be seen as that they had two sets of books parallel to each other. This was the only way to keep track of

all the financial data in the firm. This also meant that there were a lot of errors, and the accounting was done on incorrect months. This also meant that at the end of each business year, they had a lot to do. The accounting department needed to redo everything and recheck everything to make sure that it was correct. The largest cost driver was marketing spend. So, she suggested that Flygresor.se should start to closely track the spending and also the outcome of said cost. This also started the first iteration of the dashboard. Which collected marketing data via API connection directly from the platforms. By using this data combined with the sales data. It is now possible to create projections and prognoses around how the business i going. This also helped with the cleanness from the accounting side. This meant that she could see the marketing expenditure in-house instead of relying on third-party solutions.

This solutions also laid the foundation for their current dashboard. They can now use data from previous years and create a model by analysing patterns throughout the year to project earnings and sales. This is because that the travel industry work i cycles. By doing this they could regulated marketing expenditure to maximise profit and also project the return on ads spent.

"The weekly patterns were always the same, Sundays and Mondays were the best, and the coming week from the salary day was always the best in the month" - Controller

In the current state, they are within an error margin below 10% of the project outcome of the month versus the actual outcome. This model allows the CEO to see daily how close or far the firm is from its targets. All of their financial decisions are made based on this dashboard and earnings model. But this model and process are constantly tested and validated each day and iterated. This solution also eliminated the massive Excel sheet, which would break every other week because of the sheer amount of data it contains.

This solution was possible to build because of the Executive Partnership Manager. Before she arrived at the firm, they had different contracts with each supplier and travel agencies, which made it impossible to correctly measure the outcome and their earnings each day. Flygresor.se earns money by selling clicks, and all of the travel agencies have different prices per click. This meant that it was cheaper for certain customers than others. They resolved this by renegotiating the partnership agreement with every single supplier. They also migrated the tracking of click-out to their side instead of the side

of the suppliers. Which meant that they would charge them based on their number instead of what the supplies claimed the click-out is. This created an equal condition for everyone along with a standardisation of the income stream. This allowed them on a minute-by-minute basis to see their earnings since every click out is now worth the same price no matter which supplier it went to.

"I'm constantly using the dashboard and checking the results from our supplier, and seeing how we can help them to improve their traffic from us. And also pushing the supplier to improve their websites." - Executive partnership manager

Everyone works based on the KPIs and monitors the firm's performance. However, none of their individual performance is measured by KPIs. There are no performance KPIs at the individual level, meaning that they are measured collectively as an organisation. Therefore, there is no individual incentive to use the dashboard, but rather it is used voluntarily.

5.1.1 Interview with the development team

The development team consists of four software developers, not counting the CTO. There is a lack of process. Development responsibilities are divided and given to each individual by the CTO, and this is done based on experience and an understanding of the employee's capabilities and learning. Since most of the developers are senior with 10+ years of experience. A key aspect mentioned by the CTO is that since they have such senior team members, it is important to balance the possibility to learn and explore while also solving mundane tasks that are less challenging. By doing so, it will keep the senior members of the team continuing to push forward and develop new skills while maintaining a level of freedom. Once the responsibility has been given, the team members are given full autonomy of the process with a deadline and clear structure regarding the deliverables. These instructions are given verbally. The Flygresor.se development team don't have a structured way of documentation regarding their processes. Each team member is their own project manager and product owner. They are allowed to operate freely and i certain cases have direct contact with consultancies. It is like having a team, but not at the same time.

”Everyone on the team understands the process of development quite well, the importance is that we are clear with our instructions[...] It is rather an advantage not needing to maintain a lot of documentation.” - CTO

On larger projects and improvements, an issue is created and posted with detailed descriptions and Q&A attached from the product owners and, in another case, from consultants. Other daily tasks or tasks with minor priority or scale are discussed verbally, and instructions are also given verbally. A big part of the way of working is the open-plan offices. During the time of this study, it can be observed that a lot of communications happen dynamically and are very sporadic. Everyone is always available and multiple discussions are happening everywhere with idea generation and getting help for solving different issues. By doing so, they have also eliminated the need for meetings.

”I think here at Flygresor.se we have a lot of freedom to see, or at least I do. I think I can find the different parts that I want to improve or what I want to work on [...]” - Senior front-end developer

”I own my work [...] and I take pride in it.” - Senior full-stack developer

While this approach of working is reflected in the entire organisation, and has contributed to the product’s rapid development and testing. It can also create difficulties for new hires and requires a certain profile to be able to fit into the culture. Since all of the employees are given significant freedom to act and make decisions can be difficult for juniors and also dependent on their need for directions and guidance. This was a pain that was expressed by the CTO during their hiring process.

However, this issue was not shared by the junior back-end developer. He started at Flygresor.se during the first year of his computer science degree and has worked there since then. Going in as a junior developer without any experience was difficult but manageable. He was given the responsibility of one of the databases i production shortly after he had started working there. This was a big task, and he was given the instructions to clean up the dataset along with checking it for errors. But he had the mindset that you can Google everything and just learn the code. So, by being self-efficient and building on the autonomy given to him, he was able to sustain the workload.

5.1.2 Key learnings

Everyone praised their organisation structure and that it helped them in their development as a firm, and also by having a quick decision structure, it allowed everyone in the firm to test their ideas quickly, sometimes on the same day or hour as they arrived. There is a strong trust and respect between the team members and the CEO allows everyone to work in the way that they want, generous work from home policies, taking half a day in the office and half day at home, no regulation regarding lunch breaks. Everything is built on mutual understanding and trust. It was also mentioned that the transition from gut-feeling and experience decision making to DDD gave a significant improvement to the understanding of the firm and also improved efficiency. Successful marketing and using ADM regarding marketing spend is a key success factor. This also required a deep understanding of the market and the business itself. Without the dashboard and the drive to build an algorithmic approach and way of thinking, the firm would not be where it is today. Being the first mover on the market helps a lot. Almost everyone who was interviewed emphasised this point, and without this factor, it would have been very hard to enter and maintain its market shares.

5.2 Comparison case

An interview was conducted with the founder of another company to use as a comparative case. The second firm is a start-up company that specialises in the optimisation of queuing and has entered its scale-up phase. The firm currently has ten (10) employees. Founded in 2020, the firm is still very young. They had quite a chaotic period in the beginning, where they were mostly just making decisions based on gut feeling and quick fixes. The organisation back then was much smaller, which allowed for agile movement and the decisions were taken quickly. They also had the issue of technical development. They needed quick iteration and testing in the market to validate their idea and understand which sector they should focus on.

"We have managed to scale the business, but not the number of employees. We have a really good team with us, but for every good hire we have to sort out four bad ones." - Co-founder

They strived for a semi-flat organisation. In the sense that every employee

gets access to everyone and all of the dashboards. There is no barrier between the management team to the rest of the employees. But this has proven to be challenging in leveraging their growth.

"All of our decisions are aimed for growth and they are based on our KPIs [...] We are extremely data driven in our approach but still want to improve and grow." - Co-Founder

This has resulted in their shift from a fully flat organisation to a hierarchical organisation. The organisation is built around several KPIs decided by the management team along with each employee. They are called the "Holy KPIs", which are the fundamental KPIs that they have to hit, and it is these KPIs that are the deciding factor for all of the remaining ones. It functions as a waterfall structure into multiple smaller streams. The Holy KPIs are the number of users, retention rate and Annual recurring revenue (ARR). For example, the CEO is responsible for the ARR and the CTO is responsible for the retention rate, while the CMO is responsible for attracting traffic to the website. Further, the KPIs are broken down to, for example, developers who are measured on the conversion rate of website visitors, back-end developers on system availability (downtime), and support staff on customer satisfaction and response time. Every employee is given one or multiple KPIs that act as their guidance. Once the KPIs have been distributed, the employees are given autonomy in their daily work. Every employee has an individual dashboard that everyone can observe.

5.2.1 Data driven culture

It is a vision from the founders that data-driven decisions and accurate measurement become ingrained as part of the DNA. In the interview, he also mentioned that they want everyone in the company to feel ownership over their KPIs and, by doing so, they can affect their daily work. By using the KPI map that they have built, they can quickly identify changes in the system. This means if they want to increase the ARR growth by 6% and they are currently at 5%, with this map, they can understand what other KPIs are lacking and need to be adjusted to be able to hit their goals. This also allows them to create scenario analyses and prognoses based on the different KPIs. By creating a culture like this and educating the employees in this way of thinking, they have already seen results. One example was during a conversation with a customer

Table 5.1: Breakdown and summary of KPI structure

Levels	**Example of KPI**	**Responsible**	**Purpose**
Holy KPIs	Number of users, retention rate, ARR	Management, CEO, CTO	Growth
Team	Conversion rate, Downtime, Task success rate	Marketing, development, Support	Breakdown of the Holy KPIs
Individual	Conversion rate, Response time	Employee	Daily operational targets

support person regarding a change in the software, and they immediately reacted with *"This will affect my KPIs negatively and I have to [...]"*. This line of thinking and ownership is already taking root in the organisation.

This has also allowed employees to raise concerns and mitigate risk before it happens by following and understanding the KPI map, and they can react and prevent certain issues. The co-founder also expressed that people feel more ownership in their decisions, which has positively affected the daily operation of the firm and increased the quality of communication. This effect was achieved by building and introducing the heart of the organisation as algorithmic, methodical and analytical. By centralising the firm around these KPI, communications became clearer and the KPI map allows the team to consistently pinpoint issues and improvements. This way of working eliminates the chaos that existed and also mitigates the lack of documentation. This systematic approach to data analytics and data-driven decision making has transformed the organisation from a reactive environment to a proactive one, where insights and actions are guided by data to achieve operational excellence, and it has become embedded in the company culture.

6. Building an algorithmic organisation

Flygresor.se inherits an extreme example for an algorithmic organisation in a high competitive market. In the meta market competition is fierce and it is impressive that they have retained market dominance for more then a decade. While Flygresor.se.se does not follow any management framework and the traditional definition of AM it has created a robust and agile organisation. It can be observed that by dissolving the entire organisation, to numbers connected to financial performance have been the key contributor to the success. This similar approach is also adopted by the comparative case by creating a KPI map connected to financial data and breaking it down to minuscule pieces of information that are digestible on an individual level. This allowed the comparative case to grow. Both cases observed similar traits and way of thinking on how to grow and manage their respective business. Flygresor.se have managed to sustain an extremely high level of autonomy for more than a decade and managed to survive COVID without a single year in losses (Flygresor.se was profitable during the entire COVID).

In the comparative case, it follows the traditional definition and way of working regarding AM with a clear hierarchy within the organisation, with a KPI map showing the employees how their tasks are affected and what KPIs are driving the development on an individual level. Observing the comparative case, it displayed the characteristics of an traditional AM organisation. An organisation with clear hierarchy, operating with a KPI map as its operating principles. Building KPI structures to measure the effectiveness of its employees. Top managements decision are data driven and based on the teams performance. By doing so eliminating human intuition. In theory, the firm could automate more tasks and adopt a more hands-off approach by setting boundaries conditions and action based interactions. By doing if the firm can eliminate middle management and focus more on the well being and performance of the daily operations. The firm can also implement automatic tracking and award systems. Similar to the Uber application, by using daily input and previous dataset to create ”nudges” to employees and pushing them

into the direction of the firms visions. This is achievable due to the high level of monitoring from the comparison case different to Flygresor.se what have no tracking on its employees at all. During the interview with the comparison case it was also mentioned that the employees compensation is also KPI based.

Both firms observed showed a high level of autonomy and flexibility. It was also shown that both firms can achieve automation level five (5) if they want. This shows that it is a matter of data structure and internal software robustness. If these exist, the firms can level up and leverage these tools to achieve an even higher level of automation. Both cases displayed the ability to scale without the need for expanding their team. These firms are scaling their algorithmics and data which can accelerate the productivity of each employee. While not conclusive, both firms believe that they could leave for a month, and the firm will still be running without any more issues.

6.1 Key constituents of an algorithmic organisation

An algorithmic organisation can be divided into multiple sections based on inductive observations made during the study; these sections can be classified as Data, Automation, User Engagement, and Social Interactions. This structure can be observed in figure 6.1. This approach differs from previous research, where most of the focus tends to be describing AM rather than the practical implication, along with how an organisation could undergo such a shift.

The building blocks of an algorithmic organisation, as observed in the two cases presented in this study. The foundation of an AM organisation is built upon its data structure and its ability to reliably deliver and process large amounts of data in real time. Building a robust data structure allows the firm to expand and scale sustainably. Once the data structure is established, it allows the firm to develop an automation process and also create process mapping, along with eliminating repetitive tasks and creating time savings and effectively improving the firm's process and productivity. These two building blocks with be the tech stack of the firm and how future algorithms can be built and developed upon.

Once the tech stack has been established and tested, the firm can continue to design for social layers, and these are User engagement and social interactions. Based on how the technical software is designed, along with the interface. The design of the interface will dictate the amount of engagement the user will have. If the user feels that it is their decision being made, and feels responsible for it. The highest layer is about the social interactions of the day-to-day and also the cultural aspect of the organisation.

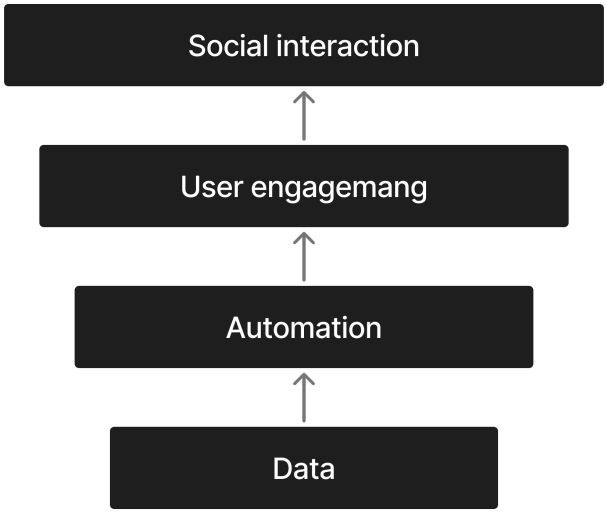


Figure 6.1: Foundation of the taxonomy with a bottom-up approach to algorithmic management.

Breaking down the taxonomy into smaller parts allows for the measurement of these components. The components are a result of the qualitative study. By creating a thematic analysis of the interviews along with the literature study, it could be derived into these components as the contributing factors of the organisation. Adopting the same line of thinking as the two cases, which has the goal of maximising the economic value gained, it can then be deconstructed into multiple contributing factors and components to the vision.

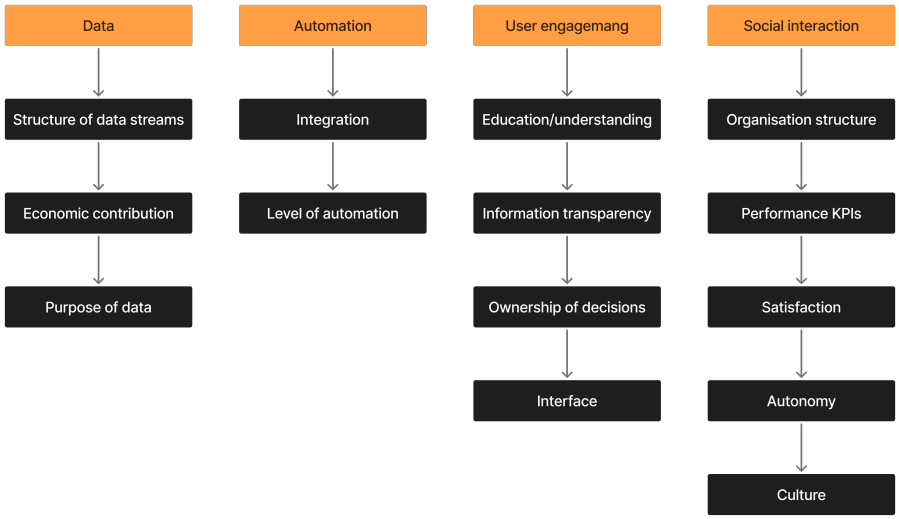


Figure 6.2: Breakdown of the components contributing to the taxonomy, to identify the level of algorithmic management adopted.

6.1.1 Data

Starting at the bottom, the foundation of an algorithmic organisation its data structure. The data segment can be divided into three key functions: structure of data streams, economic contribution and the purpose of data.

Structure of data streams. Since data is a foundation pillar of an algorithmic organisation and is dependent on the rigidity and robustness of the data pipeline, is it scalable? Can the current pipeline withstand the fluctuation of data. Is the product scalable and dynamic? This would entail a good coding practice with no hard coding. The idea of an AM organisation won't work if it breaks because of a new update.

Economic contribution. Once the data pipeline has been established, the second issue is what kind of data should be used? Looking at the case of Flygresor.se, there is an emphasis on economic contribution in the tracking of the data. This means does this data contribute to the economic growth of the firm? Can this be tracked upstream to allow the firm to gain a better understanding of its economic performance? Similarly to the comparison

case, where they developed a KPI map with three holy KPIs to live by and then attributed the remaining KPIs in a downstream approach, allowing the team to contribute and affect the firm's numbers. This approach is the other way around. For example, measuring web traffic is an important data point since this can allow for the calculation of conversion rate based on how many conversions there are in the system versus the number of clicks. This can then be calculated for the rate of conversions. Thus, contributing to the economic value and understanding of the business.

Purpose of data. Every data point should have a purpose, and this is to eliminate waste of resources trying to track and measure everything. Only measure the necessary amount to create a more efficient data structure. This can also eliminate confusion. There is no point in having data that will never be used. So, during this process of creating the necessary data pipelines, one should ask themself why this data point is being processed and measured.

6.1.2 Automation

The level of automation and the ability of automation determine how efficiently tasks can be performed within the organisation. It is closely connected to the data structure and pipeline. The level of automation can be raised depending on the robustness of said pipelines. Thus, deciding the components of the integration (number of software) and the level of automation.

Integration. This measures the level of integration within the firm. This is measured by the number of necessary dashboards and software needed during the daily operations. How much software is needed for an employee to gain an understanding of how the business is doing? The number of software needed will differ dependent on the complexity of the business and also the sector. There is not conclusive evidence from the study on how many software a firm should have. In the case of Flygresor.se.se, they have managed to narrow it down to one internal dashboard for everything. KPIs and data stream are integrated directly into one software that collects data from the different ad platforms and also the accounting systems that are used. This requires significant development resources, but this also creates simplicity regarding how the organisation operates. While this is an extreme case, firms should arguably strive for a balance.

Level of automation. There are multiple ways to measure the level of automation within a firm. The way for measuring the KPIs should be universal throughout the firm, but the levels can vary depending on the department. Since not every department can fully automate. This framework should be developed to fit the firm in question, since there is no universal solution to this issue.

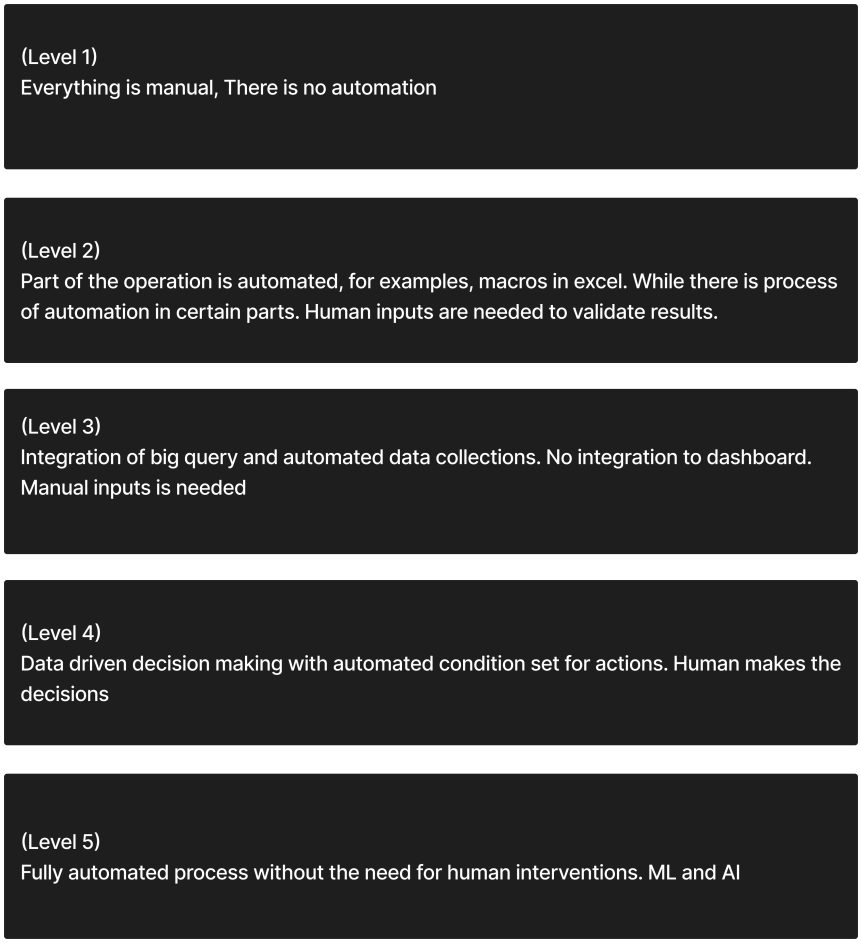


Figure 6.3: Examples of how a measuring structure can be built.

This is an example of how a firm can measure its level of automation. Adopting this model in the case of Flygresor.se.se would result in an automation level of four (4). While it is possible for Flygresor.se to reach level five (5), the firm

has no intention of doing so because the CEO have stressed the importance of owning the decisions. Using the same model for the comparison case would result in the level of three point five (3.5) since their pipeline is not fully integrated and still requires a large amount of manual work. Thus, resulting in this scoring. The management team can use this structure to analyse and create strategies based on this approach. While a high level of automation can improve efficiency and autonomy, a high level of automation does not entail a better outcome of higher profitability of the firm.

6.1.3 User engagement

Education and understanding. Previous research has shown that with AM it is sometimes like working for a black box with no transparency and understanding regarding how decisions are being made, what parameters are accountable for those decisions. In a peer-to-peer platform, the need for education regarding this might be lower than in a smaller organisation. In the case of Flygresor.se, everyone is given extensive education and explanation on how all of the KPIs and data are being collected and calculated. This is the same for the comparison case; they create the KPI map together with each employee and draw the decisions from within a collaborative approach.

Information transparency. Information asymmetry is described as a main issue within the application of AM since algorithms collect and process the data and then make the decisions, there is no transparency for the employee receiving the orders. While transparency is good, it is not always necessary, depending on the size of the firm and its operations. An overload of transparency and information can be overbearing for the employee and can cause more harm than good. By observing the case of Flygresor.se which is an algorithmic organisation with a small team, it was showed that information transparency was crucial because these teams are driven by vision. Not being transparent can lead to loss of motivation and trust, which is the foundation of an algorithmic organisation that is agile and adaptive.

Ownership of decisions. Adopting ADM can cause a lack of ownership over the decisions made within the firm. Management should strive to address the lack of ownership. The tools available should be an aid rather than a replacement for the human factor. In the case of Flygresor.se, they have the necessary measurements to aid in decisions regarding what to fix and what

to build that could aid in the development of the firm. But they engage their personnel by giving control and decision-making to their employees instead of the managers. This is also connected to information transparency, by sharing information unilaterally, this could help significantly with the inclusion and the feeling of ownership.

Interface. The interface is the integration of human and algorithms. Its design plays a key role regarding how the ownership of decision is distributed. Depending on the vision of the firm and what kind of organisation the management wants to achieve, this can be different depending on the desired outcome. If an organisation wants the user to feel high engagement with the system should be built around visualisation and aiding the user by providing the necessary information instead of creating a solution for the issue ahead. In a high engagement interface the user will be given the necessary data and tools to make an informed decision without the interception from the algorithmics. In this case the AMS only acts as support and partner in the decision making. In the case of low user engagement, the system should then prompt a full recommendation regarding what campaign should receive more or less budget, and the user will only have the choice of ”Yes” or ”No”. Which eliminated the thinking and need to act from the equation. While this can be beneficial in certain applications to increase efficiency, it can create a loss of ownership of the decision-making process, which can also lead to poor decision-making in the future (Bader and Kaiser, 2019).

6.1.4 Social interactions

In an algorithmic organisation, run on calculation and data, social interactions are at risk. Because of the level of automation and autonomy given to its employees, many might choose to work from home. This can create separation and loneliness in employees.

Organisation structure. The structure of the organisation might affect the culture of the organisation and the work process as well. While this study can not find any definitive answer regarding what type of organisational structure is most suitable for an algorithmic organisation. Top management should take this issue into account.

Satisfaction. The level of satisfaction of an employee can affect performance

and how long they stay at the firm. This issue was presented by the CTO regarding the senior developer within the team. For them to be engaged and perform on a high level, the CTO needs to balance the amount of ”boring” work versus challenging work that opens for new learning. Thus, also increases the level of satisfaction. High employee turnover can hurt a firm's ability to retain talent and create knowledge gaps.

Autonomy. In the case of Flygresor.se and the comparison case, both firms allow for a high level of autonomy for their employees. Especially Flygresor.se allowing for multiple days working from home and even half-day in the office and half-day at home set-ups. This level of autonomy is mentioned multiple times during the qualitative research, and many attributed their satisfaction and performance to the autonomy. Allowing employees to freely explore and discover new features and projects while finding new issues and discovering their abilities.

Culture. As observed from the qualitative study, culture is a key factor within an algorithmic organisation. Because of the level of autonomy given to the employees and the usage of data-driven decision making. Different cultures and affect the organisation positively or negatively. In the case of Flygresor.se, the culture is almost family-like; everyone knows exactly what to do and when to do it. No strict management, and employees are given flexible working hours and locations. This has created a way of familiarity within the firm. Employees have mentioned that a key success factor of Flygresor.se is its culture and working environment.

Performance KPIs. Measurability is a key factor in an algorithmic-driven organisation, and KPIs in general are an important aspect. Performance KPIs on an individual level are not always necessary.

7. Discussion

This study found that by leveraging algorithmic management, the firm's growth does not always mean increased headcount. It was rather the opposite. It is about the quality of the employees and the ability to retain and develop talent that becomes the main focus. This is prevalent in the case of Flygresor.se. They have not hired a full-time employee since 2019 and have still managed to grow year on year, along with retaining a good margin. The issue of hiring was also prevalent in the comparison case. Because they adopted AM without knowing it, they had issues of retaining talent and also educating them in their way of working. This will be a recurring theme in AM organisation that personal development and education will go hand in hand in the development of the firm.

7.1 Key technical factors of an algorithmic organisation

Focusing on the practical implications and challenges of an algorithmic organisation, and found no simple way to complete such a transformation. It is a complex issue regarding multiple sub-contexts. Observing what key technical and organisation factors enable successful transformation of start-ups into algorithmic-driven organisations, one of the key factors is the drive from management and the acute understanding of the need to digitalise and become data-driven. This drive is prevalent in both cases and how they communicate this throughout the organisations. Both firms have the idea of data-driven decision-making that every decision should be made based on facts and information rather than just experience and intuition. This way of thinking does not exist in all organisations. A key success factor is that their employees understand this and live by this. This is their DNA in the firm. Everyone acts and thinks in this manner. This is an important aspect. If employees do not agree or understand this work process, it would hinder the progression of the organisation. While doing something like this can take years, this was also mentioned by the comparison case, that it took them

multiple years to communicate this message, and now the employees are finally adopting it. One of the key issues with AM organisations is that it can seem scary, as if the algorithm will eliminate their jobs, leaving them jobless and that machines will take over everything.

Transforming an organisation from a start-up into a scale-up or a traditional and stable company requires alignment from management. In the case of Flygresor.se, the new CEO expressed the importance of keeping the start-up culture/vibe while the parent company would be seen as a traditional multinational corporate organisation with strict guidelines and rules. While maintaining the existing structure and culture, it allowed them to build upon this into the culture that exists today. But this was only possible because the management understood that a familiar culture would be more beneficial than a hierarchical one. While this type of culture can help with relationship building, it can also cause issues regarding productivity. This was expressed by the comparison case where they had issues with their start-up culture in that new employees weren't taking the firm seriously and were just there to "hang out," which created a lack of progress, and people weren't working. The key difference between these two cases is that the Flygresor.se team mainly consists of senior members with 10+ years of experience, and the comparison case consisted mostly of young professionals and recently graduated students. This issue requires more research to result in a conclusive result, but these are the early-stage observations. This balance of managing young talent versus experienced talent is central in an AM organisation because of the level of autonomy given to each employee. Rather than having repetitive processes, every employee will be powered by AMS to make better decisions faster than ever. This also requires alignment with the firm's vision.

Management shouldn't always stress this transformation and the expansion of its workforce. To properly implement AM, the management needs to have a good understanding of their core business and also their technical limitations. Neither of the interviewed cases did what they did overnight. It took them years to complete such a transformation. The firm needs to breathe AM and consider it as part of its DNA and not just some new methodology to adopt. Because working with AM will change the organisation structure, information flow and operational tasks. Full commitment is beneficial and often necessary to fully gain the benefits of using AM. While it might be tempting to automate everything and just allow the algorithms to take over, or just use a machine learning model to eliminate all human interactions. One should not forget

the human interactions and the human side of decision-making in the holistic view of the business. Businesses are built by humans for humans. Do not forget culture and building a high-performance and close-knit team.

7.2 Constraints, limitations and difficulties associated with algorithmic organisation

AM systems are heavily dependent on data quality and data streams, so it will be the case of garbage in and garbage out. The algorithm doesn't understand what it is calculating. This sets the stress on the input data, and also the filtering and elimination of unnecessary data points and storage. AM regarding data handling can be related to the ideology of lean manufacturing. It is the same principle of just-in-time and just-in-need. Its goal is to eliminate waste and reduce technical costs and maintenance. Reduce waste of every kind and allow the firm to only focus on the core business.

Building an AM organisation is challenging and requires a complete revamp of the organisation structure and chart. While AM is a powerful tool for data-driven companies to level up to the next level within their data handling and decision making, AM is not a miracle solution that solves everything. Not all firms and businesses can adopt AM. In the case of Flygresor.se, they have a very simple business of selling clicks, same with Uber, where they are also just selling a service and acting as a peer-to-peer partner. In the case of manufacturing companies that mainly deal with B2B sales, data is not easily documented and collected on a large scale. AM can prove to be deficient in these areas. While this does not mean that these businesses can't adopt AM, it will just mean that they will have more challenges in setting up the necessary technical solutions. In the case of firms with high technical complexity in their product offering, AM can be limiting since it may lack the understanding and the data to give proper recommendations, or it will be too complex to build such systems and also implement these solutions at every location.

7.3 Shift in the study of algorithmic management

The area of AM research is still in its infant stage. There is a lack of research in this field, with the rising demand of AI research, many research papers about AM content as AI, which creates a difficulty in differentiating these from each other. Past research tends to focus on the definition of AM and also on its effect on human behaviour, such as its affect on worker conditions mentioned by Wood, Graham, et al. (2019). Both Wood and Lehdonvirta (2021) and Möhlmann et al. (2021) spoke about information asymmetry is a key talking point across multiple sources and its effect on the employees and how firms can create ”nudges” to push employees to a certain direction. Others focus on the policy making and how policy can affect the development of AM and also its effect on society. While these issues are important, the application of AM is already deployed in multiple areas of people's daily lives, with well-known cases such as Uber, Google, Facebook and many more. Dynamic pricing is also a key aspect of AM usage, and in particular how it can improve margins and sales numbers in consumer brands. While this adoption is not new and has been heavily researched, it is still a key point in the context of AM. The term AM was first used by Lee et al. (2015) and described it as a systematic approach to personnel management. This definition and the alteration of this definition are continued in the literature. It was seen as a solution to manage a large number of employees in more efficient ways with automatic reporting, behaviour monitoring, task management and automatic onboarding and recruitment. This focus was mainly on human resources and management while missing out on its possibilities at an organisational level and using AM to increase profitability and growth.

7.3.1 Algorithmic management research within human resources

AM application within HR systems is already approaching mass adoption, even if the HR personnel don't understand or consider themselves as users of AMS. Parent-Rocheleau and Parker (2022) spoke about how these systems can reduce and also remove middle management and by doing so sliming down an organisation. By adopting this approach has yielded reduced time for HR professionals and more effective handling of employee applications

and monitoring. Allowing the managers to eliminate repetitive tasks such as creating Excel macros, reporting and tracking performance KPIs. These processes with the help of AM can now be fully automated, thus freeing up more time for the managers. The literature also explores the change in knowledge requirements for managers in AM organisations. While traditional organisations require managers with long project experience and the know-how on performance models, creating reports and documentation, in an AM organisation, these tasks will be automated and eliminated by algorithms. This also means that the managers need to know the socio-technical aspects of the team and product.

The current literature explores the potential of AM usage and its limitations of fallbacks within the peer-to-peer platform, along with the danger of using AM. Policy making precautions along with regulation can also help the development of AM on a larger scale, but it misses the practical implications of AM. The literature focuses more on the discussion and justification of AM. Missing from the literature is how one can build an AM organisation and how a firm transforms from a traditional firm into an algorithm-driven firm. In the current literature, most adoption of AM has been within managing a large amount of workforce and is seen as a tool to manage large firms with high employee turnover. While missing the application of AM in smaller firms with a low headcount. But still can leverage AM technology and the way of thinking to accelerate their development and growth. This study aims to contribute to exploring the implementation of AM in firms with smaller teams, building on the same theory and way of working to create a more efficient and profitable organisation.

7.4 Scaling an algorithmic organisation

This study resulted in one case study of Flygresor.se and their way of working, and how a small team of eight can manage a 130 million SEK a year business and still maintain profitability during tough times. The case study found that Flygresor.se was an early adopter of AM thinking and technology. They did not know about what they were doing, but had a vision of an automated business. Combining the findings from the comparison case, their path onto the usage of AM has been out of necessity since they needed to create a more effective organisation and at the same time also maintain the growth along with profitability. Scaling an AM organisation is different from scaling a

traditional organisation. It differs in the strategy regarding new recruitments and the number of employees. In an AM organisation, the company scales the software and the algorithms, and maintaining talent is a key factor in this type of organisation. An AM organisation works in symbiosis towards the same vision without anyone telling the individual what to do and when to do it. It just naturally happens. This occurrence is not necessarily random; it happens because of the transparency given to the employees. Key targets are set by the management, and everyone is given full autonomy with the only contingency that action should be taken towards reaching the vision. To summarise, scaling an AM organisation is not about head count; it's about talent and scaling the software along with its data capabilities. If successful, one can build an organisation that doesn't require constant human interactions and can, in theory, run autonomously without human input for a certain amount of time until the algorithms fail due to external factors.

7.5 Limitations

Because of the time and scope of the thesis only one comparison case was taken into account. It would have been beneficial to interview a range of companies to create a large sample size and thus, also allowing for variation in the study. Because of the limited knowledge and also the timeline it was difficult to create a more nuanced view of the study.

8. Conclusions and Future work

8.1 Conclusions

Key technical and organisational factors are a robust data pipeline and a clear understanding of what data should be included and what data should be excluded. The filtration of data should be done with a profitability mindset, focusing on measuring causes of conversion and sales, eliminating everything that does not contribute to this. Have a clear vision and organisational structure. Communicate clearly to the entire organisation regarding the implementation of AM and what that entails. Create workshops, educational material and processes for everyone. Eliminate the number of meetings and software; strive to build an easy-to-access platform or dashboard that everyone has access to that gives a clear indication of how the business is going. Eliminate information asymmetry and promote transparency and flexibility. Incorporate the idea of AM and data-driven decision making into the firm's DNA and make sure everyone understands this messaging and how to use the necessary tools. Allow for high autonomy and flexibility, allowing the employees to freely explore while working towards the vision of the firm. This approach is goal-oriented rather than process-oriented. Giving the employees freedom, education, and the necessary algorithmic tools to build and explore creates the feeling of ownership over the decisions powered by data and analytics.

The key constraints of AM are the need to build large data pipelines and infrastructures, and it can be challenging for an existing business to adapt and rebuild existing structures. This is also a difficulty associated with AM solutions that some firms might not be able to achieve because of the complexity involved. AM organisation has immense potential and can be the next step in organisation management. While challenging to implement and create, if proven successful, the return could be significant and allow firms to significantly outperform their competitor, increase their dynamic capabilities and resist unforeseen events.

8.2 Future work

For future work, more case studies should be conducted and analysed to find a practical framework that can be adopted. There are also the aspects of moral and policy making that should be explored and tested regarding how an AM organisation would be on a macro level. This study only managed to conduct two case studies of one meta search company and one subscription-based company. It would be interesting analyse cases that are in different businesses and sectors in order to gain a deep understanding of the complexity of the subject.

References

Bader, Verena and Stephan Kaiser (Sept. 2019). "Algorithmic decision-making? The user interface and its role for human involvement in decisions supported by artificial intelligence." en. In: *Organization* 26.5. Publisher: SAGE Publications Ltd, pp. 655–672. ISSN: 1350-5084. DOI: 10.1177/1350508419855714. URL: https://doi.org/10.1177/1350508419855714 (visited on 02/20/2025).

Dolgui, Alexandre and Jean-Marie Proth (Apr. 2010). "Pricing strategies and models." In: *Annual Reviews in Control* 34.1, pp. 101–110. ISSN: 1367-5788. DOI: 10.1016/j.arcontrol.2010.02.005. URL: https://www.sciencedirect.com/science/article/pii/S1367578810000118 (visited on 03/10/2025).

Duggan, James, Ultan Sherman, Ronan Carbery, and Anthony McDonnell (2020). "Algorithmic management and app-work in the gig economy: A research agenda for employment relations and HRM." en. In: *Human Resource Management Journal* 30.1, pp. 114–132. ISSN: 1748-8583. DOI: 10.1111/1748-8583.12258. URL: https://onlinelibrary.wiley.com/doi/abs/10.1111/1748-8583.12258 (visited on 02/14/2025).

Gallego, Guillermo and Garrett van Ryzin (Aug. 1994). "Optimal Dynamic Pricing of Inventories with Stochastic Demand over Finite Horizons." eng. In: *Management Science* 40.8. Publisher: INFORMS: Institute for Operations Research, pp. 999–1020. ISSN: 0025-1909. DOI: 10.1287/mnsc.40.8.999. URL: https://research.ebsco.com/linkprocessor/plink?id=4d9b6862-8c3c-398c-bec7-d8d593be686b (visited on 02/18/2025).

Gerring, John (2016). *Case Study Research: Principles and Practices (2nd ed.)* Cambridge University Press.

Günther, Wendy Arianne, Mohammad H. Rezazade Mehrizi, Marleen Huysman, and Frans Feldberg (Sept. 2017). "Debating big data: A literature review on realizing value from big data." In: *The Journal of Strategic Information Systems* 26.3, pp. 191–209. ISSN: 0963-8687. DOI: 10.1016/j.jsis.2017.07.003. URL: https://www.sciencedirect.com/science/article/pii/S0963868717302615 (visited on 03/18/2025).

Jarrahi, Mohammad Hossein, Gemma Newlands, Min Kyung Lee, Christine T. Wolf, Eliscia Kinder, and Will Sutherland (July 2021). "Algorithmic management in a work context." en. In: *Big Data & Society* 8.2. Publisher: SAGE Publications Ltd, p. 20539517211020332. ISSN: 2053-9517. DOI: 10.1177/20539517211020332. URL: https://doi.org/10.1177/20539517211020332 (visited on 02/19/2025).

Klein, Gary A. and John Sullivan (Jan. 2001). "*Sources of Power: How People Make Decisions*." EN. In: *Leadership and Management in Engineering* 1.1. Publisher: American Society of Civil Engineers, pp. 21–21. ISSN: 1532-6748. DOI: 10.1061/(ASCE)1532-6748(2001)1:1(21). URL: https://ascelibrary.org/doi/10.1061/%28ASCE%291532-6748%282001%291%3A1%2821%29 (visited on 02/27/2025).

LaValle, Steve, Michael S. Hopkins, Eric Lesser, Rebecca Shockley, and Nina Kruschwitz (Oct. 2010). "Analytics: The New Path to Value." en-US. In: *MIT Sloan Management Review*. URL: https://sloanreview.mit.edu/projects/analytics-the-new-path-to-value/ (visited on 01/24/2025).

Lee, Min Kyung, Daniel Kusbit, Evan Metsky, and Laura Dabbish (Apr. 2015). "Working with Machines: The Impact of Algorithmic and Data-Driven Management on Human Workers." In: *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*. CHI '15. New York, NY, USA: Association for Computing Machinery, pp. 1603–1612. ISBN: 978-1-4503-3145-6. DOI: 10.1145/2702123.2702548. URL: https://dl.acm.org/doi/10.1145/2702123.2702548 (visited on 02/06/2025).

Littlewood, Ken (Apr. 2005). "Special Issue Papers: Forecasting and control of passenger bookings." en. In: *Journal of Revenue and Pricing Management* 4.2. Company: Palgrave Distributor: Palgrave Institution: Palgrave Label: Palgrave Number: 2 Publisher: Palgrave Macmillan UK, pp. 111–123. ISSN: 1477-657X. DOI: 10.1057/palgrave.rpm.5170134. URL: https://link.springer.com/article/10.1057/palgrave.rpm.5170134 (visited on 02/18/2025).

Mateescu, Alexandra and Aiha Nguyen (2019). "Explainer: Algorithmic Management in the Workplace." eng. In: *SSRN Electronic Journal*. ISSN: 1556-5068. DOI: 10.2139/ssrn.4584505.

Möhlmann, Mareike, Lior Zalmanson, Ola Henfridsson, and Robert Wayne Gregory (Dec. 2021). "ALGORITHMIC MANAGEMENT OF WORK ON ONLINE LABOR PLATFORMS: WHEN MATCHING MEETS

CONTROL." eng. In: *MIS Quarterly* 45.4. Publisher: MIS Quarterly, pp. 1999–2022. ISSN: 0276-7783. DOI: 10.25300/MISQ/2021/15333. URL: https://research.ebsco.com/linkprocessor/plink?id=4eb2bdd4-62de-3061-9dc9-cf535a2fd80e (visited on 02/14/2025).

Onsrud, Harlan and James Campbell (Mar. 2020). "Being Human in an Algorithmically Controlled World." In: *International Journal of Humanities and Arts Computing* 14.1-2. Publisher: Edinburgh University Press, pp. 235–252. ISSN: 1753-8548. DOI: 10.3366/ijhac.2020.0254. URL: https://www.euppublishing.com/doi/10.3366/ijhac.2020.0254 (visited on 02/28/2025).

Parent-Rocheleau, Xavier and Sharon K. Parker (Sept. 2022). "Algorithms as work designers: How algorithmic management influences the design of jobs." In: *Human Resource Management Review* 32.3, p. 100838. ISSN: 1053-4822. DOI: 10.1016/j.hrmr.2021.100838. URL: https://www.sciencedirect.com/science/article/pii/S1053482221000176 (visited on 02/17/2025).

Phillips-Wren, Gloria and Angela Hoskisson (Jan. 2015). "An analytical journey towards big data." In: *Journal of Decision Systems* 24.1. Publisher: Taylor & Francis _eprint: https://doi.org/10.1080/12460125.2015.994333, pp. 87–102. ISSN: 1246-0125. DOI: 10.1080/12460125.2015.994333. URL: https://doi.org/10.1080/12460125.2015.994333 (visited on 01/27/2025).

Pisano, Gary, Amy Shuen, and David Teece (1997). "Dynamic capabilities and strategic management." en. In: *Strategic Management Journal* 18.7, pp. 509–533. ISSN: 1097-0266. DOI: 10.1002/(SICI)1097-0266(199708)18:7<509::AID-SMJ882>3.0.CO;2-Z. URL: https://onlinelibrary.wiley.com/doi/abs/10.1002/%28SICI%291097-0266%28199708%2918%3A7%3C509%3A%3AAID-SMJ882%3E3.0.CO%3B2-Z (visited on 03/06/2025).

Schildt, Henri (Jan. 2017). "Big data and organizational design – the brave new world of algorithmic management and computer augmented transparency." In: *Innovation* 19.1, pp. 23–30. ISSN: 1447-9338. DOI: 10.1080/14479338.2016.1252043. URL: https://doi.org/10.1080/14479338.2016.1252043 (visited on 02/07/2025).

Seele, Peter, Claus Dierksmeier, Reto Hofstetter, and Mario D. Schultz (May 2021). "Mapping the Ethicality of Algorithmic Pricing: A Review of Dynamic and Personalized Pricing." eng. In: *Journal of Business Ethics* 170.4. Publisher: Springer Nature, pp. 697–719. ISSN: 0167-4544. DOI: 1

0.1007/s10551-019-04371-w. URL: https://research.ebsco.com/linkprocessor/plink?id=b49f8f23-4140-32c2-8335-508dd6b3529d (visited on 03/07/2025).

Tomprou, Maria and Min Kyung Lee (Jan. 2022). "Employment relationships in algorithmic management: A psychological contract perspective." In: *Computers in Human Behavior* 126, p. 106997. ISSN: 0747-5632. DOI: 10.1016/j.chb.2021.106997. URL: https://www.sciencedirect.com/science/article/pii/S0747563221003204 (visited on 02/19/2025).

Wood, Alex J, Mark Graham, Vili Lehdonvirta, and Isis Hjorth (Feb. 2019). "Good Gig, Bad Gig: Autonomy and Algorithmic Control in the Global Gig Economy." en. In: *Work, Employment and Society* 33.1. Publisher: SAGE Publications Ltd, pp. 56–75. ISSN: 0950-0170. DOI: 10.1177/0950017018785616. URL: https://doi.org/10.1177/0950017018785616 (visited on 02/14/2025).

Wood, Alex J and Vili Lehdonvirta (2021). "Antagonism beyond employment: How the 'subordinated agency' of labour platforms generates conflict in the remote gig economy." In: *Socio-Economic Review* 19.4, pp. 1369–1396. DOI: 10.1093/ser/mwab016.

A. List of interview candidates

Table A.1: Interview candidates

Positions	**Process & Documendation**	**Number of interviews**
Chief executive officer	Audio recording, in-person	3
Chief Technology officer	Audio recording and in-person	3
Senior full-stack developer	Audio recording, in-person	1
Junior Software developer	Audio recording, in person	1
Senior front-end developer	Audio recording, in person	1
Hardware manager	Audio recording, in person	1
Partnership manager	Audio recording, in person	1
Financial consultant and controller	Audio recording, in person	2
Chief operations officer	Audio recording, in person	1
Chief executive officer (Firm 2)	In-person	1
	Total amount of interviews:	15

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